

ACTL2102 - Estimation

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Likelihood Function

Likelihood Function

The likelihood of transitions is given by

$$\mathcal{L} = \prod_i \prod_{j \neq i} (\lambda_{ij}^{n_{ij}} e^{-\lambda_{ij} T_i})$$

where $i \in S$, n_{ij} is the number of transitions from i to j , T_i is the total time spent in i , and λ_{ij} is the instantaneous rate from i to j .

Individual Contribution

Consider a *HSD* model. Denote μ to be the force of mortality for healthy, σ the transition rate from healthy to sick, ρ the recovery rate for the sick, and ν the force of mortality for sick. Then, for each life ℓ , the likelihood contribution L_ℓ is:

$$L_\ell = e^{-T_{\ell H}(\mu+\sigma)} \mu^{n_{\ell HD}} \sigma^{n_{\ell HS}} e^{-T_{\ell S}(\nu+\rho)} \nu^{n_{\ell SD}} \rho^{n_{\ell SH}}$$

where $T_{\ell i}$ is the time the life spends in state i , and $n_{\ell ij}$ is the number of times the life ℓ transitions from state i to j .

MLE Estimate

Maximising the likelihood for a specific parameter, we have

$$\hat{q}_{ij} = \frac{\sum_{\ell} n_{\ell ij}}{\sum_{\ell} T_{\ell i}}.$$

Very intuitive! Number of events / number of opportunities for events = rate.

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Since these are MLE's they have asymptotic distribution,

$$\hat{q}_{ij} \sim \mathcal{N} \left(q_{ij}, \frac{q_{ij}}{\mathbb{E}[T_i]} \right) \sim \mathcal{N} \left(q_{ij}, \frac{q_{ij}^2}{\mathbb{E}[N_{ij}]} \right).$$

Thank you for the term!

Good luck with your finals :) I hope to tutor you again in future courses!

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